

; Write X86 menu driven Assembly Language Program (ALP) to implement OS (DOS) commands TYPE, COPY and DELETE using file operations.
; User is supposed to provide command line arguments in all cases.

```
%include "macro.asm" ; Include macro file (assumed to
define fopen, fread, fwrite, fclose, fcreate, fdelete)
```

```
section .data
    dispMsg db "Write X86 menu driven Assembly
Language Program (ALP) to implement OS (DOS) commands
TYPE, COPY and DELETE using file operations. User is
supposed to provide command line arguments in all
cases.",10,"Name:- Sumaanyu",10,"Roll:- 7256",10,"Date of
Performance:- 17/04/2025",10
    dispMsglen equ $-dispMsg
```

```
    msg db "-----MENU-----", 10
    db "    1. TYPE ", 10
    db "    2. COPY ", 10
    db "    3. DELETE ", 10
    db "    4. Exit ", 10
    db "Enter your choice : "
    msglen equ $-msg
```

```
    endl db 10
    m db "DONE!", 0ah ; Message to display when
action is complete
```

```
section .bss
    choice resb 2 ; Store user menu choice
    fname1 resb 50 ; First filename (source)
    fname2 resb 50 ; Second filename (destination
or target)
```

```
    filehandle1 resq 1 ; File handle for source
    filehandle2 resq 1 ; File handle for destination
```

```
    buffer resb 100 ; Buffer to hold file contents
    bufferlen resb 1 ; Actual length of data read
    temp resb 16 ; Temporary storage (not used
```

here)

```
section .text
    global _start
```

```
_start :
    pop rbx ; Number of command-line
arguments
    pop rsi ; Program name (skip it)
    write dispMsg, dispMsglen ; Display description
header
```

```
    ; Read file names from command-line arguments
    mov rdi, fname1 ; Destination buffer for first file
```

mark :

```
    pop rsi ; Get next argument (filename)
    mov rdx, 0
```

next :

```
    mov al, byte[rsi+rdx] ; Copy character
```

```
    mov [rdi+rdx], al
    cmp byte[rsi+rdx], 0 ; End of string?
    je next1
    inc rdx
    jmp next
```

next1 :

```
    cmp rdi, fname2 ; If first filename copied, go for
```

second

```
    je main_menu
    mov rdi, fname2
    jmp mark
```

; ----- Main Menu -----

main_menu :

```
    write msg, msglen
    read choice, 2
```

```
    cmp byte[choice], "1"
    je case1
```

```
    cmp byte[choice], "2"
    je case2
```

```
    cmp byte[choice], "3"
    je case3
```

```
    cmp byte[choice], "4"
    je case4
```

```
    jmp main_menu
```

; ----- TYPE command -----

case1 :

```
    call type
    jmp main_menu
```

; ----- COPY command -----

case2 :

```
    call copy
    jmp main_menu
```

; ----- DELETE command -----

case3 :

```
    call delete
    jmp main_menu
```

; ----- Exit -----

case4 :

```
    mov rax, 60 ; syscall: exit
    mov rdi, 0
    syscall
```

; ----- TYPE FUNCTION -----

type :

```
    ; Open source file in read mode
    fopen fname1
    cmp rax, -1h
    je case4 ; Exit if file open fails
```

```
    mov [filehandle1], rax
```

```

; Read file contents into buffer
fread [filehandle1], buffer, 100
dec rax          ; Adjust length (optional)
mov [bufferlen], rax

write fname1, 50    ; Show filename
write endl, 1
write buffer, bufferlen ; Show file contents
write endl, 1

fclose fname1      ; Close file
ret

```

; ----- COPY FUNCTION -----

copy :

```

; Open source file
fopen fname1
cmp rax, -1h
je case4
mov [filehandle1], rax

; Read file contents
fread [filehandle1], buffer, 1000
mov [bufferlen], rax

; Create destination file
fcreate fname2
cmp rax, -1h
je case4
mov [filehandle2], rax

; Write contents to destination file
fwrite [filehandle2], buffer, [bufferlen]

; Close both files
fclose fname1
fclose fname2

write m, 6    ; Print DONE!
ret

```

; ----- DELETE FUNCTION -----

delete :

```

fdelete fname2 ; Delete destination file
write m, 6    ; Print DONE!
ret

```